Spacekime Analytics:
A Novel Framework for Time Series Modeling and fMRI Data Analysis

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1 Introduction of the General Idea

The project is driven by our interest in understanding how discrete music categories are represented in human brain. Inspired by Nakai T's work **"Correspondence of categorical and feature-based representations of music in the human brain"** published in 2021. We focused on the analysis of high-dimensional fMRI data by first testing out the performance of benchmark models for music classification task (traditional statistical models and machine learning models) and introducing a novel framework for time series analysis which is particularly useful in analyzing fMRI data.

2 New Framework for Time Series Analysis: Kime Algorithm

2.1 Model Assumption

Consider a time series with length K, N repeated measurements. According to the Kime Algorithm, we assume that for each observation, there exists a latent angle variable $\theta_{j,k}$ behind each time step with range $[-\pi, \pi]$. Thus the model pre-settings takes the form of:

$$Y_{j,k} = S(t_k, \theta_{j,k}) + \epsilon_{j,k} = a_0(t_k) + \sum_{n=1}^{N_{harm}} [a_n(t_k)\cos(n\theta_{j,k}) + b_n(t_k)\sin(n\theta_{j,k})] + \epsilon_{j,k}$$

- j = 1, 2, ..., N (N individuals/ repeated measurements)
- k = 1, 2, ..., K (K time steps)
- $\epsilon_{j,k} \sim N(0, \sigma^2)$
- We represent $S(t, \theta)$ by a finite-order fourier expansion in θ. Specifically, the series is truncated at order N_{harm}, corresponding to N_{harm} harmonic coefficients.

Goal Estimate $a_n(t_k), b_n(t_k), \phi^{(k)}$

2.2 Initialization

- N_{harm}: Decide the number of harmonic coefficients.
- a_0, a_n, b_n: Setting a_0, a_n, b_n to 0.
- σ^2: Randomly set a value for σ^2.
- ϕ: Initialize as uniform distribution.

2.3 E Step

Weight Calculation To compute the posterior weights, we first define a fixed grid of phase values $\{\theta_g\}_{g=1}^{G}$.

For each observation i at time step t_k, the posterior weight on this discrete mesh is defined as:

$$w_{i,g}^{(k)} = Pr(\theta = \theta_g | Y_{i,k})) = \frac{p(Y_{i,k}|\theta_g)\phi_g^{(k)}}{\sum_{g'=1}^{G} p(Y_{i,k}|\theta_{g'})\phi_{g'}^{(k)}}$$

where $\{\phi_g^{(k)}\}_{g=1}^{G}$ represents the predicted discrete phase distribution at time t_k. Conditioned on θ_g and t_k, the distribution of $Y_{i,k}$ is:

$$Y_{i,k}|\theta_g, t_k \sim N(S(t_k, \theta_g), \sigma^2)$$

so that:

$$\log p(Y_{i,k}|\theta_g, t_k) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(Y_{i,k} - S(t_k, \theta_g))^2}{2\sigma^2}$$

For numerical stability, all computations are carried out in the log domain. We define

$$s_{i,g}^{(k)} := \log(p(Y_{i,k}|\theta_g)\phi_g^{(k)}) = \log p(Y_{i,k}|\theta_g) + \log \phi_g^{(k)}$$

and compute the normalized weights as

$$w_{i,g}^{(k)} = \frac{\exp(s_{i,g}^{(k)})}{\sum_{h=1}^{G} \exp(s_{i,h}^{(k)})} = \exp(s_{i,g}^{(k)} - LSE_i^{(k)})$$

where the log-sum-exponential normalization term is $LSE_i^{(k)} = \log \sum_{h=1}^{G} \exp(s_{i,h}^{(k)})$.

2.4 M step

a_n, b_n **Update** We construct a finite Fourier basis expansion over the angular variable θ_g as

$$B_g = \begin{bmatrix} 1 \\ \cos(1\theta_g) \\ ... \\ \cos(H\theta_g) \\ \sin(1\theta_g) \\ ... \\ \sin(H\theta_g) \end{bmatrix} \in R^p$$

and stack all G basis vectors to form the design matrix

$$B = \begin{bmatrix} B_1^T \\ \vdots \\ B_G^T \end{bmatrix} \in R^{G \times p}$$

The corresponding coefficient vector is defined as

$$\beta = [a_0, a_1, ..., a_H, b_1, ..., b_H]^T$$

For each time point t_k, the coefficients $\beta^{(k)}$ are estimated by solving a ridge-regularized weighted least squares problem:

$$\hat{\beta}^{(k)} = \arg\min_{\beta} \sum_{i=1}^{N} \sum_{g=1}^{G} w_{i,g}(Y_{i,k} - B(\theta_g)^T\beta)^2 + \lambda||\beta||_2^2$$

The closed form solution takes the form of:

$$\hat{\beta}^{(k)} = (B^T diag(r^{(k)})B + \lambda R)^{-1} B^T t^{(k)}$$

where $R = I_p$ with $R_{11} = 0$, and $r_g^{(k)} = \sum_{i=1}^{N} w_{i,g}^{(k)}$, $t_g^{(k)} = \sum_{i=1}^{N} w_{i,g}^{(k)} Y_{i,k}$.

ϕ **Update** For each time point t_k, let $W^{(k)} = [w_{i,g}^{(k)}]_{i=1:N,g=1:G}$ denote the individual-level posterior weight matrix. We first compute the marginal posterior mass at each grid point θ_g by summing over individuals:

$$m_g^{(k)} = \sum_{i=1}^{N} W_{i,g}^{(k)}, g = 1, ..., G$$

The raw discrete distribution of θ is then obtained by normalizing these marginals:

$$\phi_g^{(k),raw} = \frac{m_g^{(k)}}{\sum_{h=1}^{G} m_h^{(k)}}, \sum_{g=1}^{G} \phi_g^{(k),raw} = 1$$

In the case that all $m_g^{(k)} = 0$, we initialize $\phi^{(k),raw}$ as a uniform distribution:

$$\phi_g^{(k),raw} = \frac{1}{G}, \forall g$$

Then we use gaussian kernel or other kernel functions smoothing strategy to get our smoothed estimate of ϕ:

$$\phi_g^{(k),smooth} = \frac{\sum_{h=1}^{G} \phi_h^{(k),raw} K(\frac{\theta_g - \theta_h}{bw})}{\sum_{g'=1}^{G} \sum_{h=1}^{G} \phi_h^{(k),raw} K(\frac{\theta_{g'} - \theta_h}{bw})}$$

where $K(\cdot)$ is gaussian kernel

$$K(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

bw is bandwidth.

To further control the noise, we apply damped update for ϕ:

$$\phi^{(k),new} = (1 - \alpha)\phi^{(k),old} + \alpha\phi^{(k),smooth}$$

where $\alpha \in (0, 1]$ is the damping factor.

Comments on σ^2 σ^2 here performs as the temperature parameter in the calculation of weight. According to previous part, we have:

$$w_{i,g}^{(k)} \propto p(Y_{i,k}|\theta_g)\phi_g^{(k)}$$

with $Y_{i,k}|\theta_g \sim N(S(t_k, \theta_g), \sigma^2)$. Thus we know:

$$p(Y_{i,k}|\theta_g) \propto \exp(-\frac{(Y_{i,k} - S(t_k, \theta_g))^2}{2\sigma^2})$$

From the above expression, σ^2 actually controls how sharply the posterior weights are concentrated over the θ_g grid, a smaller σ produces very peaked weights (favoring θ_g values close to $Y_{i,k}$) while a larger σ gives more spread-out smoother weights.

2.5 Reconstruct Surface

After iteration, the revered 2d surface can be expressed as:

$$\hat{S}(t_k, \theta_g) = a_0(t_k) + \sum_{n=1}^{n_{harm}} \hat{a}_n(t_k)\cos(h\theta_g) + \hat{b}_n(t_k)\sin(h\theta_g)$$

$$g = 1, .., G, k = 1, ..., K$$

Another output of the algorithm is the estimated discrete distribution of latent variable ϕ, which can perform as a useful tool to generate 1-d signals.

2.6 Algorithm Summary

Algorithm 1: KPT

Input: n time series of length k: $\mathbf{X} = \{x_1, ..., x_n\}, x_i \in R^k$, theta grid

Output: Reconstructed surface $\hat{S}(t_k, \theta_g)$ and estimated distribution of θ at each time point $\hat{\phi}^{(k)}$

Step 1: Initialize $N_{harm}, \sigma^2, a_n, b_n, \phi$.

Step 2: **while** *not converged* **do**

- **E step:**
- Calculate weight for each θ_g at each time point;
- **M Step:**
- Update a_n, b_n;
- Update $\phi_g^{(k)}$;

3 Simulation Result

3.1 Data Generation

We generate 100 time series with 5 time points, set the number of harmonic coefficient to $3(N_{harm} = 3)$, assume the true harmonic coefficients as:

$$a_n(t) = \begin{cases} \cos(\pi n t), & n \in \{2, 3\} \\ 5\cos(\pi t), & n = 1 \end{cases}$$

$$b_n(t) = \begin{cases} 0.5\sin(0.2\pi(n+1)t), & n \in \{2, 3\} \\ 2\sin(0.2\pi(n+1)t), & n = 1 \end{cases}$$

$$t_k = (k-1)\Delta t$$

$$\Delta t = 0.4, k = 1, 2, 3, 4, 5$$

Note we enlarge the coefficient at $n = 1$, this is just for better visualization. For simplicity, suppose θ at each time point has the same wrapped laplace distribution:

$$\theta_{j,k} \sim Wrapped\ Laplace(\mu, b)$$

with $\mu = \frac{\pi}{2}, b = 1$, we extract θ samples for each observation at each time point from this distribution.

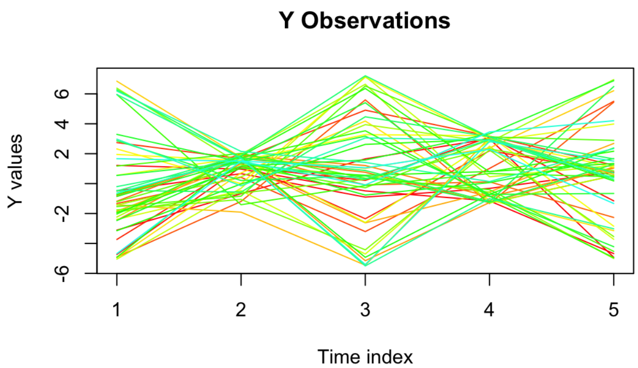

Figure 1: Simulated Signals with length = 5

Then, we use the following formula to generate samples:

$$Y_{j,k} = \sum_{n=1}^{N_{harm}} [a_n(t_k)\cos(n\theta_{j,k}) + b_n(t_k)\sin(n\theta_{j,k})] + \epsilon_{j,k}$$

where $\epsilon_{j,k} \sim N(0, \sigma^2)$, $\sigma = 0.02$, again for simplicity, we do not include the intercept term. The true surface is shown as below:

3.2 Initialization

- $N_{harm} = 3$.
- $\sigma = 0.1$.
- $a_n(t_k) = 0, b_n(t_k) = 0$.
- Set theta grid as 20 points evenly located on $[-\pi, \pi]$:

$$\theta_g = -\pi + (g-1)\Delta\theta \in [-\pi, \pi]$$

$$\Delta\theta = \frac{\pi}{10}$$

- $\lambda = 0.01$
- $\alpha = 0.3$
- $bw = 0.1$
- max iteration $= 3000$

3.3 Results

Converged at iteration $= 2400$.

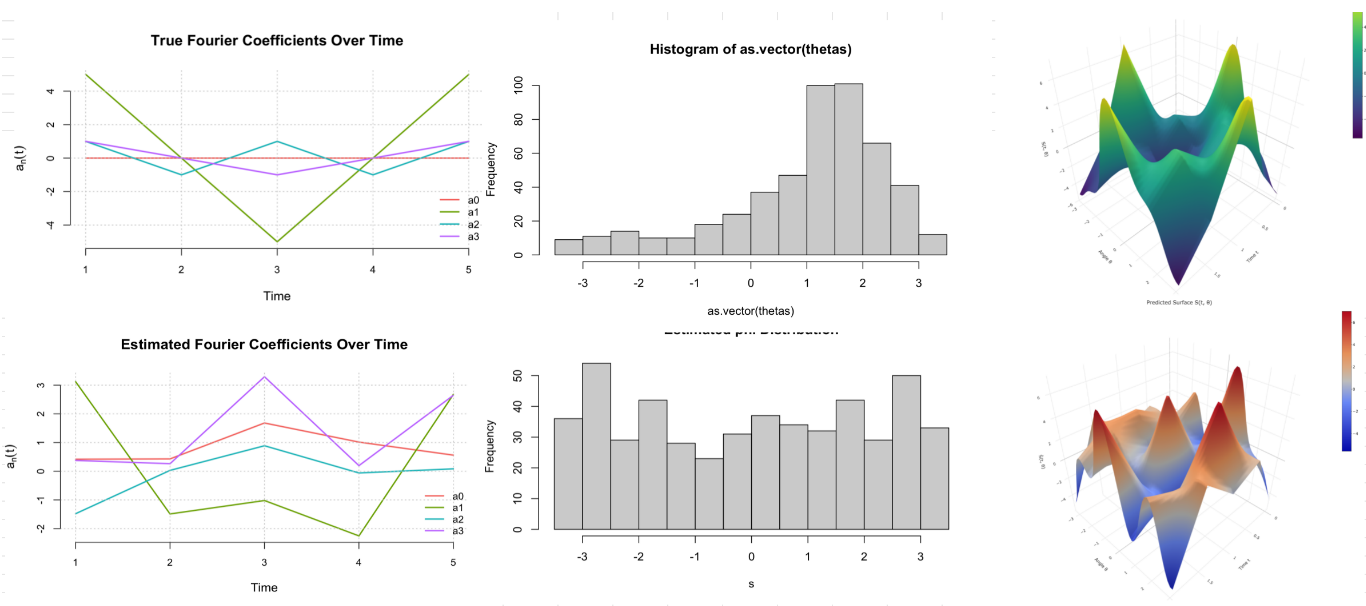

Figure 2: Recovered Result, first line is the true surface features, the second line is the recovered surface features

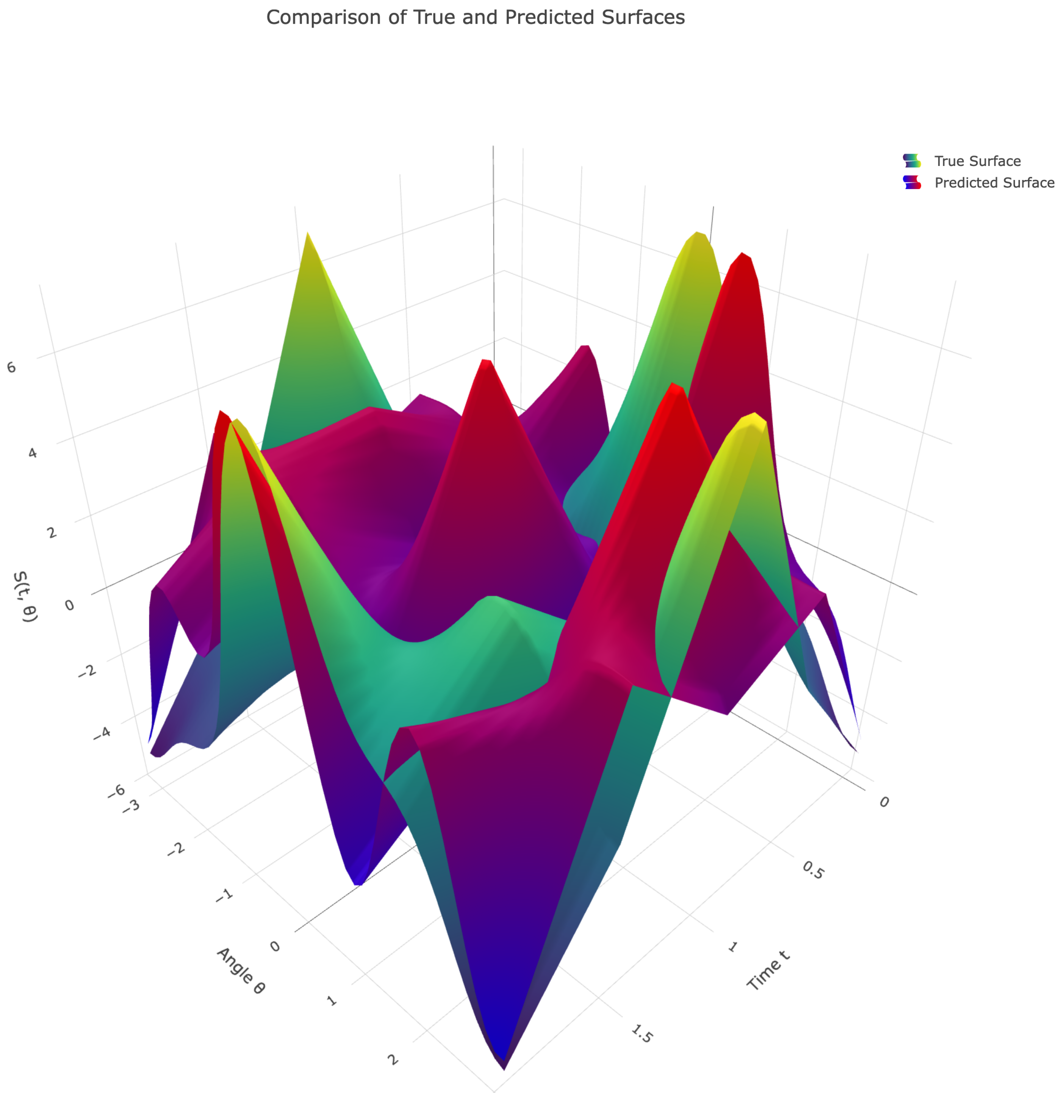

Figure 3: True Surface and Recovered Surface

Loglikelihood

$$\hat{S}(t_k, \theta_g) = \hat{a_0}(t_k) + \sum_{n=1}^{N_{harm}} [\hat{a}_n(t_k)\cos(n\theta_g) + \hat{b}_n(n\theta_g)\sin(n\theta_g)]$$

$$\log L = \sum_{k=1}^{K}\sum_{i=1}^{N}\log\sum_{g=1}^{G}\exp(-\frac{1}{2}\log(2\pi\sigma^2) - \frac{(Y_{i,k} - \hat{S})^2}{2\sigma^2} + \log\hat{\phi}(\theta_g|t_k))$$

MSE mse = 11.712

4 Follow-up Question

- How to choose number of harmonic coefficient?
- Decide σ^2?
- May need more validation for model assumption.
- Require more experience on how to finetuning the model.

- Require more efficient way to update phi.